A DIGITAL MARKETPLACE FOR UPCYCLING AND VALORIZING SRI LANKA'S TEXTILE WASTE

INDIVIDUAL THESIS

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Member 03 — Traceability, AI Pricing & Buyer Recommendation

Group ID: [Group ID]

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

After their initial use, materials found in textile waste may still have economic and circular value. However, fragmented information on fabric type, condition, quantity, location, demand, and market context might make it challenging to determine an appropriate commercial value and find prospective buyers. Member 03 creates digital traceability and the Commercial Intelligence layer in TexCycle AI. The component maintains a digital lifespan through traceability events, rates possible buyer groups based on material and market compatibility, and determines a reference textile price using structured marketplace and textile attributes.

On the supplied assessment dataset, the reported price-prediction experiment yields an MAE of 12.24, RMSE of 15.40, and R2 of 0.9590 using an XGBoost regression technique. In the present prototype evaluation, buyer suggestion indicates a matching accuracy of 83.56% using Random Forest and XGBoost techniques for compatibility rating. Textile recyclers, apparel manufacturers, fashion designers, furniture makers, insulators, cleaning cloth manufacturers, exporters, and toy manufacturers are among the possible buyer categories. Textile analysis, listing, offer, transaction, delivery, impact, and associated lifecycle events are all connected via traceability.

The findings show prototype-level technological viability. Because the marketplace dataset utilized for the prediction tests is artificial, they cannot demonstrate real-market accuracy. Real Sri Lankan textile prices, buyer interactions, transaction histories, and industry input are needed for future validation. Therefore, rather than a claim of production-grade market forecast, the main contribution is an integrated, localized commercial intelligence and traceability workflow.

Keywords: circular economy, textile marketplace, AI pricing, buyer suggestion, traceability, commercial intelligence, textile waste, and XGBoost

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I also thank the entire TexCycle AI team for their effort. While Member 04 marketplace analytics provide supplementary demand and marketplace predictions, Member 01 Fabric Intelligence and Member 02 Sustainability Intelligence outputs offer crucial upstream information for business choices. The duties allocated to Member 03 are the exclusive subject of this thesis.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
F1	Harmonic mean of precision and recall
FastAPI	Python framework for APIs
LKR	Sri Lankan Rupee
ML	Machine Learning
RF	Random Forest
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
XGB	XGBoost
XGBoost	Extreme Gradient Boosting
UI	User Interface

1 INTRODUCTION

1.1 Project background

Through production, processing, distribution, and consumption, Sri Lanka's textile and garment industry contributes significantly to the country's economy [1], [2]. Therefore, textile waste poses a burden to the environment as well as a chance to recover value through recycling, repair, reuse, and other circular solutions [3], [4]. Material identity is not the only factor that determines a textile waste batch's worth. Its commercial usefulness can be significantly impacted by conditions, quantity, location, demand, competition, environmental factors, and prospective buyers [4], [5], [9].

By handling textile waste as a structured digital circular inventory, TexCycle AI overcomes this difficulty. Material and condition features are identified by Fabric Intelligence; environmental and circular indicators are estimated by Sustainability Intelligence; pricing estimation, buyer recommendations, and lifecycle traceability are added by Member 03; and broader marketplace forecast is added by Member 04. Therefore, the Member 03 contribution serves as a link between commercial market activity and textile properties evaluated by AI.

1.2 Member 03 problem context

Sellers may be aware of a textile batch's approximate composition and quantity, but they are still unsure of what constitutes a fair reference price and which buyer group is most likely to utilize it. Before determining if a listing is appropriate, buyers also want information on material attributes, quantity, condition, sustainability, and seller context. Furthermore, a circular marketplace needs proof of a textile's journey from research to listing, offer, transaction, and delivery. A component that combines commercial intelligence and traceability is motivated by these criteria.

1.3 Problem statement

The lack of an integrated decision-support layer that can rank compatible buyer categories, estimate a reference textile price, and keep an organized lifecycle history for textile marketplace activities is the specific issue that Member 03 addresses. It must deal with structured outputs generated by other AI modules and link those outputs to the workflows of buyers and sellers.

1.4 Aim

The goal is to create, deploy, and assess an integrated Commercial Intelligence and Traceability component that facilitates lifecycle event monitoring, buyer-category suggestion, and AI-assisted textile pricing calculation inside the TexCycle AI marketplace.

1.5 Objectives

• Using pertinent textile and marketplace information, create an AI-assisted reference-price prediction pipeline.

• Use MAE, RMSE, and R2 to assess the price model.

• Create a buyer-recommendation system that prioritizes buyer groups based on market and textile suitability.

• Use the available prototype metric to assess buyer matching.

• Create a traceability system that covers significant events in the textile lifecycle.

• Combine seller marketplace procedures with commercial outputs.

• Offer organized data that listings, offers, transactions, and downstream marketplace services may use.

• Keep track of restrictions and avoid portraying prototype metrics as real-world assurances.

1.6 Research questions

• RQ1: To what extent may marketplace attributes and structured textiles aid in reference-price estimation?

• RQ2: How well can a textile listing's buyer categories be ranked by the system?

• RQ3: How can a consistent history of marketplace activities pertaining to textiles be preserved through digital traceability?

• RQ4: Without closely tying the AI models to the marketplace user interface, how can pricing, buyer suggestion, and traceability be integrated?

• RQ5: How does the present synthetic marketplace dataset affect validation?

1.7 Scope

Digital lifecycle traceability, buyer-category suggestion, and reference-price calculation are all included in this component. The scope include buyer-side suggestion planning and development as well as seller-side marketplace functionality that has already been developed and documented. Although those outputs could be used by the larger marketplace process, it does not assert ownership of Member 04's listing-success, sales-time, demand, or market-trend models.

1.8 Individual contribution

Area	Member 03 responsibility	Output
AI pricing	Feature preparation, price model, evaluation and API integration	Reference price
Buyer recommendation	Buyer categories, compatibility features, model integration	Ranked buyer suggestions
Traceability	Lifecycle events and history structure	Traceability event history
Seller workflow	Price/buyer intelligence integration	Commercial listing support
Research	Literature, methodology, evaluation and limitations	Individual thesis evidence

1.9 Thesis structure

The research problem and Member 03 scope are presented in Chapter 1. Related work is reviewed in Chapter 2. The research approach is presented in Chapter 3. System and component design is covered in Chapter 4. Implementation and integration are explained in Chapter 5. Results and validation are presented in Chapter 6. Interpretation and restrictions are covered in Chapter 7. The individual contribution is concluded and further work is identified in Chapter 8. Supporting information is provided in the appendices, glossary, and references.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Textile circular economy

The goal of circular economy methods is to decrease reliance on virgin resources while maintaining product and material value [3], [11]. Reuse, recycling, and circular supply-chain techniques are identified as crucial paths in textile waste management research, which also highlights stakeholder, technological, and financial obstacles [4]. This implies that in a marketplace, material and environmental attributes should be taken into account in addition to commercial worth.

2.2 AI-assisted textile value assessment

Textile categorization and condition evaluation can be aided by computer vision and machine learning [12], [13], [17]. These results produce structured qualities that are transferable to commercial models downstream. However, market value is not only determined by image-based categorization. Additional factors that affect pricing include quantity, condition, location, demand, competition, market expansion, and economic environment [4], [14].

2.3 Price prediction

When the objective is continuous, like a reference price, machine-learning regression works well. After proper preprocessing, tree-based boosting techniques may handle mixed feature interactions and simulate nonlinear connections among structured features. For the price-prediction challenge, the project description chooses XGBoost and assesses it using MAE, RMSE, and R2.

2.4 Recommendation systems

Recommendation systems leverage compatibility between attributes to rate potential goods or users. Instead of a general consumer-item recommendation problem, the task in the TexCycle setting is buyer-category recommendation. When rating buyer categories, the project takes into account fabric type, quantity, condition, customer category, location, price, sustainability norms, and demand [9], [14].

2.5 Digital traceability

Throughout an item's existence, digital traceability keeps organized records of important occurrences. Research on Digital Product Passports offers a helpful conceptual foundation for maintaining structured product data and lifecycle history [7], [8]. This idea is used to the analysis, listing, offer, transaction, delivery, effect, and buyer rating of textile marketplace events.

2.6 Research gap

Existing direction	Strength	Remaining limitation	TexCycle response
Textile AI sorting	Material recognition	Does not provide marketplace valuation	Commercial layer consumes textile AI outputs
Price prediction	Continuous valuation	Often ignores textile circularity	Price model uses textile + market features
Recommendation systems	Candidate ranking	Not textile-waste specific	Buyer categories tied to textile use
Digital Product Passports	Lifecycle information	Not necessarily AI marketplace integrated	Traceability events connected to marketplace
Circular marketplaces	Buyer-seller connection	Limited integrated AI decision support	AI analysis → sustainability → commercial → marketplace

2.7 Conceptual contribution

Integration and Sri Lankan localization are the contributions. The initiative makes no claims on the novelty of price prediction, advice, or traceability on its own. Rather, Member 03 combines these skills to the results of Fabric Intelligence and Sustainability Intelligence by integrating them around a textile-waste listing.

3 RESEARCH METHODOLOGY

3.1 Research design

The component combines experimental machine-learning assessment with an applied Design Science Research focus. Problem identification and literature study are followed by data preparation, model selection, training, assessment, artifact preservation, API integration, and application testing [4], [9].

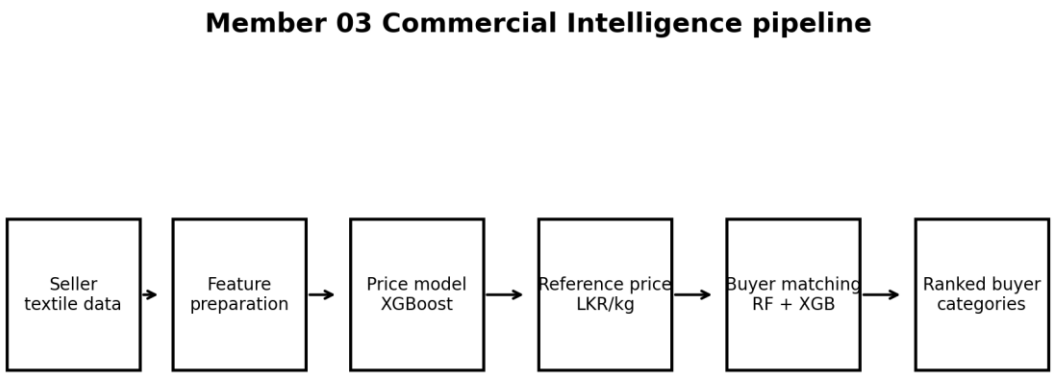


Figure 3.1 Member 03 Commercial Intelligence pipeline

3.2 Research inputs

Textile properties and an organized marketplace are used in the pricing process. Fabric type, condition, quantity/weight, location, demand, competition, market growth, average market price, seller attributes, and economic indicators are among the reported project feature categories [4], [14]. Material compatibility and buyer criteria, such as fabric type, quantity, condition, client category, location, price, sustainability requirements, and market demand, are used in buyer recommendations.

Feature group	Examples	Use
Textile	Fabric type, condition, weight	Price + buyer compatibility
Location	District, province	Local market context
Demand	Demand level / market signals	Price + buyer relevance
Competition	Competition indicators	Price context
Market	Growth, average market price	Price context
Seller	Seller qualities / experience	Commercial context
Sustainability	Sustainability / green indicators	Buyer suitability
Buyer criteria	Material category, quantity, location	Recommendation ranking

3.3 Dataset strategy

There are currently 100,000 artificial records with 36 variables in the marketplace prediction dataset. It is not a representative sample of real Sri Lankan transactions, but it is appropriate for illustrating the technological procedure. Strong metrics on synthetic data do not demonstrate production market accuracy, hence this difference is upheld throughout the thesis [9], [10].

3.4 Data preprocessing

Numerical variables that are structured are examined and converted into a format that is suitable for models. Learned preprocessing processes are used to encode categorical variables. The outlined methodology includes feature engineering, data cleaning, exploratory analysis, and the separation of training and assessment data. To avoid train/inference mismatch, the same fitted preprocessing logic must be applied again during inference.

3.5 Price model methodology

The goal is the reference textile price, which is represented in the defined commercial workflow as an output in the form of a price per kilogram. An XGBoost regressor is the model. Rather than being an immediately binding transaction price, the output is meant to serve as a reference value. Actual bids, logistics, quality assurance, and market negotiations may still be taken into account by the vendor.

3.6 Buyer recommendation methodology

Buyer-category specifications and textile characteristics are included in the buyer suggestion. The Random Forest and XGBoost models are mentioned in the published implementation. Rather of guaranteeing that a particular buyer would acquire the listing, the recommendation result is a sorted collection of potential buyer groups.

3.7 Traceability methodology

Structured entities and events that link textile data and business activities are used to implement traceability. Analysis, listing, offer, transaction, delivery, and effect are all included in the defined lifecycle. The broader process also includes buyer rating. In the real implementation, every event should have a timestamp, actor/reference, and pertinent object IDs.

3.8 Evaluation metrics

Task	**Metric**	**Purpose**
Price regression	MAE	Average absolute prediction error
Price regression	RMSE	Penalizes larger prediction errors
Price regression	R^2	Explained variance / fit indicator
Buyer recommendation	Accuracy / matching accuracy	Overall correct matching rate in current prototype
Traceability	Functional completeness	Whether required lifecycle events can be recorded and retrieved
Integration	API success/error rate	Reliability of component communication

3.9 Responsible interpretation

An estimate produced using the presumptions reflected in the training data is called a price prediction. A buyer recommendation does not constitute a transaction guaranty; rather, it is a compatibility suggestion. A traceability record is proof of a digital event that was recorded; it is not proof that the underlying physical activity took place. For reliable deployment, these distinctions are crucial.

4 PRODUCT AND SOLUTION DESIGN

4.1 System requirements

ID	Requirement	Type	Acceptance evidence
CI-01	Generate reference price from structured textile inputs	Functional	Price value returned
CI-02	Expose price prediction to seller workflow	Functional	UI/API integration
CI-03	Rank compatible buyer categories	Functional	Ordered recommendations
CI-04	Use relevant textile and market attributes	Functional	Feature contract
TR-01	Record textile analysis event	Functional	Traceability record
TR-02	Record listing event	Functional	Traceability record
TR-03	Record offer event	Functional	Traceability record
TR-04	Record transaction event	Functional	Traceability record
TR-05	Record delivery/impact events where available	Functional	Lifecycle history
NFR-01	Handle invalid input safely	Non-functional	Controlled error
NFR-02	Maintain model version traceability	Non-functional	Version metadata
NFR-03	Separate AI service from application backend	Non-functional	FastAPI/Node boundary

4.2 Commercial Intelligence pipeline

Marketplace services and upstream intelligence provide Member 03 with structured characteristics. While the suggestion pipeline assesses compatibility with potential buyer groups, the pricing pipeline converts these features into a reference price. The seller workflow may then access both outputs and link them to a listing.

4.3 Traceability lifecycle

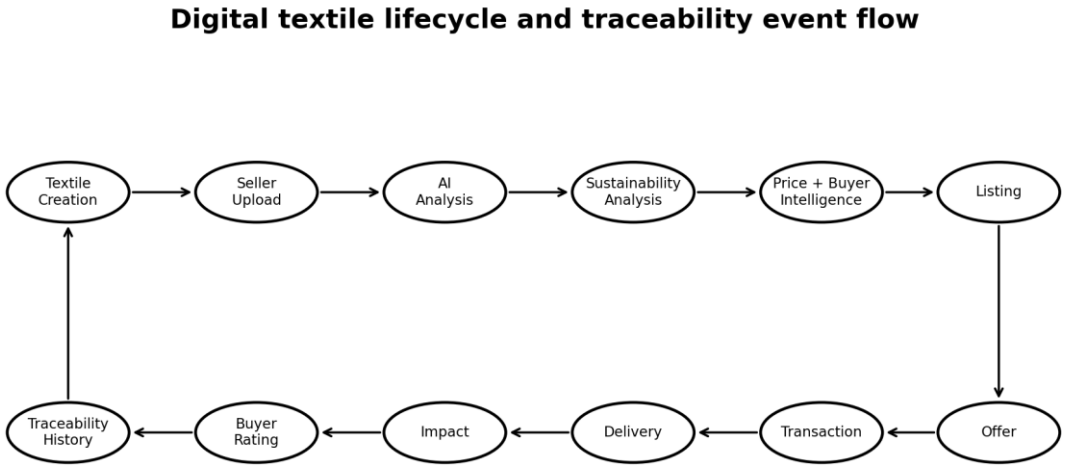


Figure 4.1 Digital textile lifecycle and traceability event flow

From the first textile or record creation to seller upload, AI analysis, sustainability analysis, commercial intelligence, listing, offer, transaction, delivery, impact, and buyer rating, the traceability concept maintains a digital history. Future Digital Product Passport-style enhancements and auditability are supported by the event sequence [7], [8].

4.4 Component architecture

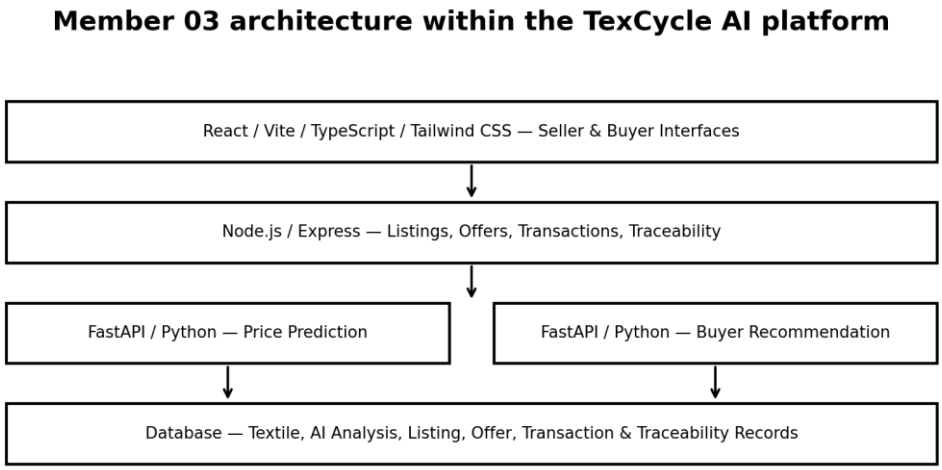


Figure 4.2 Member 03 architecture within the TexCycle AI platform

The design divides the database-backed lifecycle records, Python/FastAPI AI services, Node.js/Express application layer, and React/Vite display layer. This division makes it possible to replace or retrain the pricing and recommendation algorithms without having to completely rework the marketplace user interface.

4.5 Price-prediction design

Stage	Input/output	Design decision
Input	Textile + market attributes	Structured feature contract
Preprocessing	Categorical/numerical transformation	Fitted preprocessing reused at inference
Model	XGBoost Regressor	Nonlinear tabular prediction
Output	Reference price	Seller decision support
Persistence	Prediction + model metadata	Traceability/reproducibility
Integration	Seller listing workflow	Price visible before listing

4.6 Buyer-recommendation design

Compatibility is used to rank the types of potential buyers. Textile recyclers, apparel manufacturers, fashion designers, furniture manufacturers, insulation makers, cleaning cloth manufacturers, exporters, and toy manufacturers are among the recorded categories. Material kind, quantity, condition, client category, location, price, sustainability norms, and market demand can all be included in the ranking [9], [14].

4.7 Traceability event design

Event	Purpose	Representative information
Analysis	Record AI assessment	Textile ID, analysis reference, timestamp
Listing	Record marketplace publication	Listing ID, seller, price, timestamp
Offer	Record buyer interest	Buyer/category, offer, timestamp
Transaction	Record accepted commercial action	Transaction reference, parties, value
Delivery	Record fulfillment	Delivery state, timestamp
Impact	Record circular outcome	Impact reference / status
Rating	Record post-transaction feedback	Rating, reviewer, timestamp

4.8 Design principles

• Modularity: Marketplace business logic and AI services are kept separate.
• Traceability: Records and model versions should continue to be responsible for significant lifecycle actions and forecasts.
• Human error: rather than taking the place of negotiation, business forecasts reinforce it.
• Localization: Support is provided for LKR, Sri Lankan districts and provinces, and pertinent buyer categories.
• Reproducibility: model artifacts, evaluation outcomes, and preprocessing should all be versioned.
• Security: Authentication and authorization should be used to safeguard private seller/buyer information.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Technology stack

Layer	Technology	Member 03 role
Frontend	React, Vite, TypeScript, Tailwind CSS	Display price, buyer suggestions and marketplace records
Backend	Node.js, Express.js	Listings, offers, transactions and traceability
ML service	Python, FastAPI	Price/recommendation inference
Price model	XGBoost Regressor	Reference price
Recommendation	Random Forest + XGBoost	Buyer compatibility
Database	MongoDB / database-backed storage	Lifecycle and commercial records
Model artifacts	Pickle/model files + encoders	Deployment/reproducibility

5.2 Model artifacts

The marketplace-intelligence artifacts price_prediction_model.pkl, demand_prediction_model.pkl, buyer_recommendation_rf.pkl, buyer_recommendation_xgb.pkl, and buyer_label_encoder.pkl are specifically mentioned in the project settings. Demand, listing-success, sales-time, and market-trend artifacts are part of the larger marketplace intelligence scope and are not claimed here as Member 03 research ownership; instead, Member 03 concentrates on the price and buyer-recommendation artifacts.

5.3 Price prediction implementation

Before producing the reference price, the implementation loads the fitted pricing model and runs the associated feature-preparation process. The API boundary should handle missing or incorrect data, validate needed fields, and provide a regulated response. When possible, model loading should take place at service initialization rather than periodically for each request.

5.4 Buyer recommendation implementation

The recommendation service loads the buyer label encoder along with the fitted Random Forest and XGBoost artifacts. Buyer categories that are legible by humans are created from candidate outputs. Instead of displaying internal class indices, the user interface might display ranking categories.

5.5 Traceability implementation

Together with textile, artificial intelligence, sustainability, price forecast, buyer suggestion, listing, offer, transaction, and market snapshot data, traceability is preserved. The event structure should be created in a way that prevents the historical analysis used for a previous listing from being erased by subsequent modifications to pricing or recommendation models.

5.6 Seller workflow

• The seller adds photos and details about textiles.
• Information on fabric and condition is provided by Fabric Intelligence.
• Sustainability Intelligence generates cyclical and environmental results.
• A reference price is generated by Member 03.
• Potential buyer categories are ranked by Member 03.
• After reviewing the commercial data, the seller edits or creates a listing.
• Additional traceability events are produced by offers and transactions.
• The lifetime record may be extended by delivery, impact, and buyer-rating events.

5.7 Re-analysis and re-grading

The project documentation distinguishes re-analysis from re-grading. Re-analysis reruns fabric/defect analysis using stored listing images and can then trigger sustainability recalculation. Re-grading recalculates sustainability outputs from stored AI analysis without necessarily rerunning the image model. For Member 03, historical price and recommendation outputs should similarly retain their model version so that later model changes do not silently overwrite past evidence.

5.8 Test plan

Test ID	Scenario	Expected result
CI-T01	Valid price-prediction input	Reference price returned
CI-T02	Missing price feature	Controlled validation error
CI-T03	Invalid categorical value	Controlled error or mapped validation
CI-T04	Price model unavailable	Graceful service error
CI-T05	Buyer recommendation request	Ranked buyer categories returned
CI-T06	Unknown buyer label	Controlled response
CI-T07	Create listing after prediction	Commercial output associated with listing
TR-T01	Create analysis event	Traceability record stored
TR-T02	Create listing event	Listing event stored
TR-T03	Create offer event	Offer event stored
TR-T04	Create transaction event	Transaction event stored
TR-T05	Retrieve lifecycle history	Chronological events displayed
TR-T06	Repeat same query	Stable output under same model/version
INT-T01	Member 02 → Member 03 hand-off	Required sustainability fields available
INT-T02	Member 03 → Member 04 hand-off	Price/buyer outputs available downstream
SEC-T01	Unauthorized traceability access	Access denied
SEC-T02	Invalid payload	No server crash

5.9 Security and privacy

Sensitive commercial data includes vendor characteristics, offers, transaction prices, and buyer identities. Role-based authorization and authentication should be mandated by the marketplace. Private transaction records and uploaded material photos shouldn't be accessible via public endpoints. Environment secrets and model files should not be kept in publicly editable application locations.

6 RESULTS AND VALIDATION

6.1 Price prediction results

The documented price-prediction experiment reports MAE = 12.24, RMSE = 15.40 and $R^2 = 0.9590$ on its evaluation dataset [4], [9]. The result indicates strong fit within the available development/evaluation data. Because the marketplace dataset is

synthetic, the metric must be interpreted as prototype evidence rather than as a guaranteed market-price error in Sri Lanka.

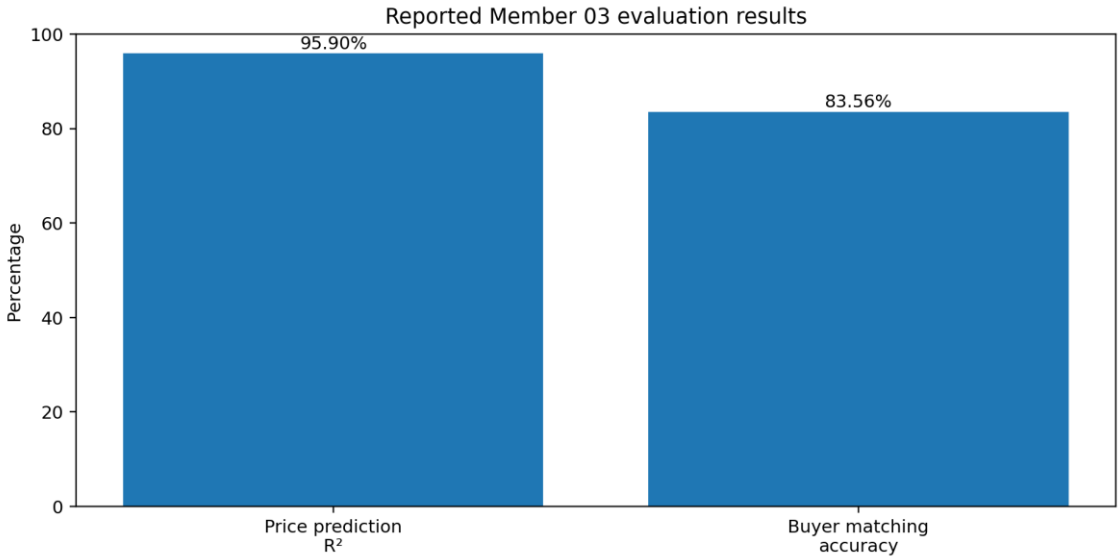


Figure 6.1 Reported Member 03 evaluation results

Model	Metric	Result	Interpretation
Price Prediction	MAE	12.24	Average absolute error on evaluation data
Price Prediction	RMSE	15.40	Error measure emphasizing larger deviations
Price Prediction	R^2	0.9590	Strong fit on documented evaluation data
Buyer Recommendation	Matching accuracy	83.56%	Current prototype matching result

6.2 Buyer recommendation result

The buyer recommendation/matching experiment reports 83.56% matching accuracy in the current prototype [9], [14]. This result should be interpreted as a model-level prototype result under the documented evaluation setting. It does not establish that 83.56% of real buyers would accept a recommendation. A stronger future evaluation should use real interaction, offer or transaction outcomes and report ranking-oriented measures where appropriate.

6.3 Price and buyer outputs in an example listing

The project documentation records an example 50-kilogram Denim Textile Waste listing. The interface example reports a predicted reference price of LKR 260.15/kg, a listed price of LKR 109.16/kg and an estimated total value of LKR 13,007.50. Textile recyclers are shown as the best buyer match in the example. These are prototype interface outputs and should not be presented as independent market observations [4], [9], [14].

6.4 Traceability validation

Traceability is validated functionally rather than through a predictive accuracy metric. The core evidence is whether events can be created, linked to the correct textile/listing/transaction records, retrieved chronologically and protected by authorization. A traceability system is only useful when the event chain remains consistent across updates, offers and transactions.

6.5 Integration validation

The Member 03 component fits into the larger workflow after Fabric and Sustainability Intelligence. The upstream modules provide material, condition and environmental attributes; Member 03 adds reference price and buyer categories; Member 04 can then consume relevant commercial attributes for marketplace prediction. This modular sequence prevents each component from independently processing the same raw image.

6.6 Result interpretation

The combination of $R^2 = 0.9590$ for price prediction and 83.56% buyer matching accuracy demonstrates technical feasibility of the prototype. It does not prove market adoption or real-world generalization. Synthetic-data bias, limited transaction history, changing prices, buyer preference variation and market drift remain major threats to validity.

6.7 Evaluation evidence requirements

- Preserve the exact train/test split used for the reported price result.

- Preserve preprocessing and model artifacts.
- Report raw prediction errors where possible.
- Report buyer recommendation confusion or ranking evidence rather than only overall accuracy.
- Separate model metrics from single-prediction confidence values.
- Record model version for every historical commercial prediction.
- Validate the traceability event sequence using actual application records.

7 DISCUSSION

7.1 Commercial intelligence value

The principal value of Member 03 is decision support at the point where a textile becomes marketplace inventory. A seller can move from material and sustainability information toward a reference value and potential buyer categories without manually switching between unrelated tools. This supports a more coherent circular marketplace workflow.

7.2 Interpretation of price performance

The high R^2 indicates that the model captured relationships present in the evaluation data. However, because the dataset is synthetic, the result may partly reflect the way the data-generation rules encode the target. The model should therefore be considered a prototype estimator until it is tested against independently collected Sri Lankan transaction prices.

7.3 Interpretation of buyer matching

The 83.56% matching accuracy provides initial evidence that the selected features and algorithms can distinguish among buyer categories in the current prototype. Nevertheless, buyer recommendation is ultimately a behavioural task. A category that appears compatible from structured attributes may not purchase due to logistics, minimum order quantities, quality requirements, timing or negotiated price. Real offer and transaction outcomes are therefore required for stronger validation.

7.4 Traceability contribution

Traceability adds a non-predictive but strategically important layer. It connects AI analysis with marketplace actions and creates a history that can later support auditing, dispute resolution, Digital Product Passport development and circular-impact reporting. The event chain also helps explain which analysis and model version were associated with a listing at the time it was created.

7.5 Integration with other members

Member	Output consumed by Member 03	Member 03 output
Member 01	Fabric type, condition, health, repairability, lifespan	Used as commercial features
Member 02	Sustainability/green/environmental indicators	Used in buyer suitability and marketplace context
Member 03	—	Price, buyer recommendation, traceability
Member 04	Marketplace/demand context	Commercial outputs can enrich predictive analytics

7.6 Strengths

- Combines pricing, recommendation and traceability in one individual component.
- Uses structured features from upstream AI rather than duplicating image processing.
- Provides a clear separation between model inference and marketplace transactions.
- Supports local currency and Sri Lankan geographic attributes.
- Maintains the possibility of model-versioned historical records.
- Provides practical seller-facing outputs before listing publication.

7.7 Limitations

Limitation	Effect	Control/future solution
Synthetic marketplace data	Weak real-world generalization	Collect real transaction data
Cold start	Limited behavioural evidence	Use controlled pilot marketplace
Price drift	Model becomes stale	Continuous monitoring/retraining
Buyer behaviour	Compatibility ≠ purchase	Validate using offers/transactions

Negotiated pricing	Predicted price may differ from final price	Show reference price, not fixed price
Regional differences	Demand varies by location	Local market features and validation
Traceability input quality	Recorded event may not prove physical action	Role verification and audit controls
Data privacy	Commercial records may be sensitive	Authorization and secure storage

7.8 Ethical and responsible AI

A commercial AI system should not create false certainty. Price predictions should be displayed as estimates, with appropriate context. Buyer recommendations should be explainable at a useful level, for example by indicating compatibility with fabric category, quantity or sustainability requirements. The system should avoid discriminatory seller/buyer profiling and should not expose private information merely because it is available in the database.

7.9 Research contribution

The research contribution is the localization and integration of AI pricing, buyer recommendation and lifecycle traceability around textile waste. This is more defensible than claiming that the individual algorithms are novel. The project demonstrates how multiple AI and information-management capabilities can be connected into a circular marketplace decision-support workflow.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

Member 03 developed the Commercial Intelligence and Traceability component of TexCycle AI. The component provides a reference price using an XGBoost regression pipeline, ranks potential buyer categories using Random Forest and XGBoost approaches, and maintains structured lifecycle events for textile marketplace activity. The documented price experiment achieved MAE 12.24, RMSE 15.40 and R^2 0.9590, while buyer matching achieved 83.56% accuracy in the current prototype.

8.2 Objective achievement

Objective	Status	Evidence
Develop price prediction	Achieved at prototype level	XGBoost price model
Evaluate price model	Achieved	MAE 12.24, RMSE 15.40, R^2 0.9590
Develop buyer recommendation	Achieved at prototype level	RF + XGB recommendation
Evaluate buyer matching	Achieved at prototype level	83.56% matching accuracy
Develop traceability	Achieved at workflow level	Lifecycle event structure
Integrate seller workflow	Implemented/documented	Price and buyer outputs linked to listing process
Real-market validation	Not yet achieved	Requires real Sri Lankan data

8.3 Future work

- Collect verified Sri Lankan textile transaction and pricing data.
- Integrate real-time market indicators and dynamic pricing.
- Use real buyer interactions, offers and transactions for recommendation evaluation.
- Introduce ranking metrics such as Precision@K, Recall@K or NDCG for buyer recommendation.
- Add buyer preference profiles with explicit consent and explainable matching.
- Expand traceability toward Digital Product Passport-style records.
- Add model-version and data-version metadata to historical predictions.
- Introduce model drift monitoring for price and buyer behaviour changes.
- Add fraud and anomaly detection for suspicious listings and transactions.
- Improve logistics-aware buyer matching using location and delivery cost.
- Support Sinhala, Tamil and English buyer/seller workflows.
- Establish independent external validation before commercial claims.

8.4 Final research statement

The current evidence supports the feasibility of an integrated commercial-intelligence and traceability workflow. The strongest next step is not simply to increase model complexity, but to improve the quality and representativeness of the data, validate the system with real industry stakeholders and preserve transparent traceability of predictions and transactions.

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GLOSSARY

Term	Definition
AI Pricing	Machine-learning-assisted estimation of a reference textile price.
Buyer Recommendation	Ranking of buyer categories according to compatibility with a textile listing.
Commercial Intelligence	Member 03 component for pricing and buyer decision support.
Traceability Event	Structured record representing a significant textile lifecycle action.
Reference Price	Estimated price used to support seller decision-making; not a guaranteed transaction price.
MAE	Mean Absolute Error.
RMSE	Root Mean Squared Error.
R^2	Coefficient of determination.
XGBoost	Extreme Gradient Boosting machine-learning framework.
Random Forest	Ensemble of decision trees used for classification/regression.
Cold Start	Limited prediction evidence caused by few historical marketplace interactions.
Model Drift	Change in prediction behaviour caused by changes in the underlying data distribution.

APPENDIX A — COMMERCIAL FEATURE DICTIONARY

Feature	Description	Member 03 use
fabric_type	Project fabric category	Price + buyer compatibility
condition	Condition category	Price + buyer suitability
weight_kg	Textile quantity	Price + buyer suitability
district	Sri Lankan district	Local market context
province	Sri Lankan province	Geographic context
demand_level	Market demand indicator	Price + buyer matching
competition	Competition indicator	Price context
market_growth	Market growth indicator	Price context
average_market_price	Market reference	Price context
seller_experience	Seller characteristic	Commercial context
sustainability_score	Environmental/circular indicator	Buyer suitability

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX B — PRICE MODEL CARD

Field	Description
Task	Reference textile price regression
Algorithm	XGBoost Regressor
Primary metrics	MAE, RMSE, R^2
Reported MAE	12.24
Reported RMSE	15.40
Reported R^2	0.9590
Intended use	Marketplace decision support
Not intended for	Guaranteed market valuation
Main limitation	Synthetic development/evaluation data

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APPENDIX C — BUYER CATEGORY SPECIFICATION

Category	Potential compatibility
Textile recycler	Materials suitable for recycling
Clothing producer	Reusable fabric/material inputs
Fashion designer	Higher-condition/design-oriented textiles
Furniture producer	Textile inputs for furniture applications
Insulation manufacturer	Suitable textile waste for insulation
Cleaning-cloth producer	Suitable lower-value absorbent material
Exporter	Marketable textile batches for external markets
Toy producer	Suitable textile inputs for toy-related production

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APPENDIX D — TRACEABILITY EVENT SCHEMA

Event	Minimum logical fields
Analysis	textile_id, analysis_id, timestamp, model_version
Listing	listing_id, textile_id, seller_id, price, timestamp
Offer	offer_id, listing_id, buyer_id/category, amount, timestamp
Transaction	transaction_id, listing_id, parties, value, timestamp
Delivery	delivery_id, transaction_id, state, timestamp
Impact	impact_id, textile_id/transaction_id, outcome, timestamp
Rating	rating_id, transaction_id, reviewer, score, timestamp

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APPENDIX E — API CONTRACT CHECKLIST

Area	Validation
Price input	Required fields validated
Price output	Numeric value and model metadata
Buyer input	Required textile/buyer features
Buyer output	Human-readable ranked categories
Traceability write	Authenticated event creation
Traceability read	Authorized lifecycle retrieval
Errors	Controlled HTTP/application responses
Versioning	Model/data version retained

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APPENDIX F — EXTENDED TEST CASES

ID	Scenario	Expected
CI-101	Valid price request	Prediction returned
CI-102	Missing weight	Controlled validation
CI-103	Unknown fabric	Controlled validation
CI-104	Extreme quantity	Safe response
CI-105	Price model missing	Service error handled
BR-101	Valid recommendation	Ranked categories
BR-102	Unknown buyer category	Controlled mapping
BR-103	No compatible buyer	Low-confidence/empty result handled
TR-101	Create listing event	Event persisted
TR-102	Create offer event	Event persisted
TR-103	Create transaction event	Event persisted
TR-104	Retrieve history	Chronological history

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APPENDIX G — ERROR TAXONOMY

Error	Impact	Mitigation
Out-of-distribution price inputs	Unreliable estimate	Validation + human review
Missing feature	Prediction failure	Input validation
Market drift	Model becomes stale	Monitoring/retraining
Buyer mismatch	Poor recommendation	Real interaction validation
Duplicate event	Incorrect history	Event IDs / idempotency
Unauthorized access	Privacy breach	RBAC

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APPENDIX H — MODEL EVALUATION PROTOCOL

Step	Evidence
1	Freeze evaluation dataset
2	Record preprocessing version
3	Load exact model artifact
4	Generate predictions
5	Calculate MAE/RMSE/R^2
6	Preserve raw prediction file
7	Evaluate buyer matching
8	Record model version
9	Interpret results against dataset limitations

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX I — BUYER RECOMMENDATION EVALUATION

Measure	Current/future evidence
Matching accuracy	Current reported 83.56%
Precision@K	Future real-user evaluation
Recall@K	Future real-user evaluation
NDCG@K	Future ranked-recommendation evaluation
Acceptance rate	Future buyer interaction study
Offer conversion	Future transaction validation

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APPENDIX J — TRACEABILITY VALIDATION PROTOCOL

Test	Pass criterion
Event creation	Correct event persisted
Event linkage	Correct textile/listing/transaction reference
Ordering	Chronological retrieval
Authorization	Only permitted roles can view/write
Update integrity	Historical event not silently overwritten
Auditability	Model version and timestamp available

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX K — SELLER WORKFLOW EVIDENCE PLAN

Evidence	Purpose
Price forecast screen	Show AI pricing output
Buyer suggestion screen	Show ranked buyer categories
Listing screen	Show commercial output connected to listing
Offer screen	Show traceability progression
Transaction screen	Show commercial event
Traceability history	Show lifecycle chain

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX L — BUYER WORKFLOW EVIDENCE PLAN

Evidence	Purpose
Search/filter	Buyer discovery
Textile details	Material/condition visibility
Sustainability details	Environmental context
Seller details	Decision context
Offer/purchase	Transaction initiation
Rating/review	Post-transaction event

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX M — SECURITY CHECKLIST

Control	Requirement
Authentication	Required for private commercial records
Authorization	Role-based access
Input validation	All API payloads validated
File safety	Uploaded content handled safely
Secrets	Not stored in source code
Logging	Avoid unnecessary sensitive data
Model protection	Artifacts not publicly writable

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX N — RESPONSIBLE AI CHECKLIST

Principle	Application
Transparency	Label price as estimate
Human oversight	Allow seller/buyer judgement
Fairness	Avoid discriminatory buyer/seller profiling
Privacy	Protect transaction and identity data
Accountability	Retain model/version metadata
Robustness	Monitor drift and failures

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APPENDIX O — CONTRIBUTION TIMELINE

Week	Member 03 documented work
1	Marketplace requirements, seller/buyer requirements, pricing/recommendation discussion
2	Textile marketplace and pricing research
3	AI pricing, recommendation and traceability literature research
4	Pricing/buyer dataset investigation
5	Marketplace/pricing data organization
6	Data cleaning and feature preparation
7	Colab training environment and baseline pricing model
8	Price model training and evaluation
9	Algorithm comparison and error analysis
10	Feature engineering and tuning
11	Feature importance and model review
12	Model preparation for integration
13	Price prediction testing and API requirements
14	Price API integration and frontend/backend testing
15	Seller marketplace-analysis functionality
16	Buyer recommendation requirements and flow
17	Recommendation development and seller testing
18	Sri Lankan apparel-market research
19	Pricing/recommendation methodology documentation
20	Buyer-side development and PP2 preparation
21	Final testing and viva evidence

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APPENDIX P — MODEL VERSIONING PROTOCOL

Item	Requirement
Model artifact	Versioned filename or registry entry
Preprocessor	Versioned with model
Feature schema	Versioned
Dataset	Version/date retained
Evaluation	Metric record retained
Inference	Model version attached to result
Historical listing	Original prediction preserved

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APPENDIX Q — DATA QUALITY PROTOCOL

Check	Purpose
Duplicate detection	Avoid leakage
Missing-value audit	Prevent hidden inference failures
Category validation	Keep consistent labels
Range checks	Detect impossible quantities/prices
Outlier review	Investigate unusual market values
Split integrity	Prevent train/test contamination

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APPENDIX R — VIVA QUESTIONS AND ANSWERS

Question	Answer focus
Why XGBoost?	Structured tabular prediction and nonlinear relationships
What does R^2 0.9590 mean?	Strong fit on evaluation data, not real-market guarantee
Why buyer recommendation?	Reduce manual buyer discovery
Is 83.56% real buyer acceptance?	No; it is current prototype matching accuracy
Why traceability?	Preserve lifecycle history and auditability
Can predicted price be final price?	No; it is a reference estimate
What is biggest limitation?	Synthetic data / limited real transactions

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APPENDIX S — REPRODUCIBILITY CHECKLIST

Item	Evidence
Dataset	Source and split
Features	Schema documented
Preprocessing	Encoder/scaler version
Model	Artifact retained
Evaluation	Metrics retained
API	Endpoint verified
Frontend	Actual workflow tested
Traceability	Event records tested
Screenshots	Real system evidence

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APPENDIX T — FINAL SUBMISSION CHECKLIST

Requirement	Status
A4 format	Completed
SLIIT margins	Completed
Times New Roman	Completed
12 pt body	Completed
1.5 spacing	Completed
Chapter heading style	Completed
Roman preliminary pagination	Configured
Arabic main pagination	Configured
Figures inserted	Completed
Tables inserted	Completed
In-text citations	Included
Reference list	Included
No fabricated UI screenshots	Complied
Actual UI screenshots	Insert from tested build if available
Student ID	IT22215338
Supervisor/group placeholders	To be completed before submission

This appendix provides supporting evidence and reproducibility guidance for the Member 03 Commercial Intelligence and Traceability component. Any implementation-specific field names should be checked against the final deployed code before submission.

APPENDIX U — RESEARCH INTEGRITY AND EVIDENCE BOUNDARY

The available project sources support the reported prototype results, architecture, model families and workflow described in this thesis. They do not support a claim that the price model has been validated against a representative Sri Lankan transaction population, nor that the buyer recommendation has been validated through a completed field study. Accordingly, this thesis explicitly separates documented prototype metrics from future validation requirements.

Actual application screenshots have not been fabricated. The technical figures inserted in this document are diagrams created to explain the implemented architecture and workflow. Before final submission, real screenshots from the tested build should be inserted where they provide stronger evidence.